


# Winning Space Race with Data Science

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22nd September, 2026

# Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

# Executive Summary

In this capstone project, It was predicted, using different machine learning classification algorithms, if the SpaceX Falcon 9 first stage will land successfully.

This project included the following processes:

- Data collection, wrangling, and formatting.
- Exploratory data analysis.
- Data visualization.
- Machine learning prediction.

Finally, it is concluded that decision tree machine learning algorithm, has the best best performance to predict if the Falcon 9 first stage will have a successful landing.

# Introduction

The commercial space age has arrived, and major companies are making space travel more affordable. SpaceX is the most successful, achieving major milestones such as sending spacecraft to the International Space Station and significantly reducing launch costs. While a SpaceX rocket launch costs around $62 million, other providers charge up to $165 million each; much of these savings stem from SpaceX's ability to reuse the first stage.



As a result, this project gathered data to build dashboards and machine learning models to answer the following questions:

- What operational factors are fundamental in determining whether a rocket will land successfully?
- Will the first stage be reused?

Section 1

# Methodology



# Methodology

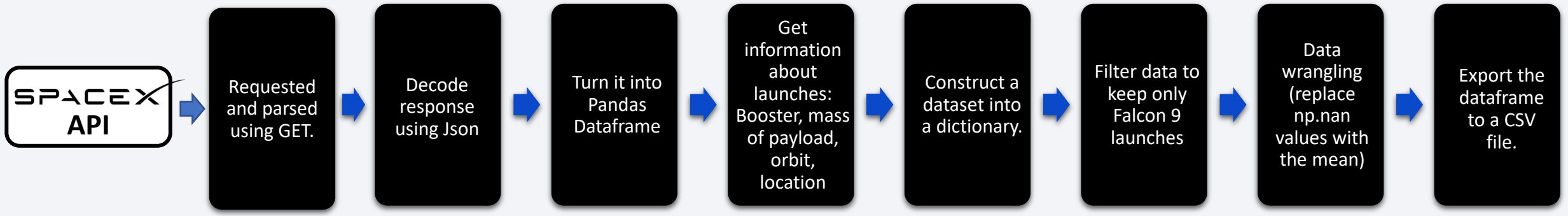
The methodology using include the following steps

- Data collection, wrangling, and formatting from the following sources:
  - ✓ SpaceX Rest API
  - ✓ (Web Scrapping) from Wikipedia
- Perform exploratory data analysis (EDA) using visualization and SQL using:
  - ✓ Matplotlib and Seaborn
  - ✓ Folium
  - ✓ Dash
- Perform predictive analysis using the following classification models:
  - ✓ Logistic regression
  - ✓ Support vector machine (SVM)
  - ✓ Decision tree
  - ✓ K-nearest neighbors (KNN)

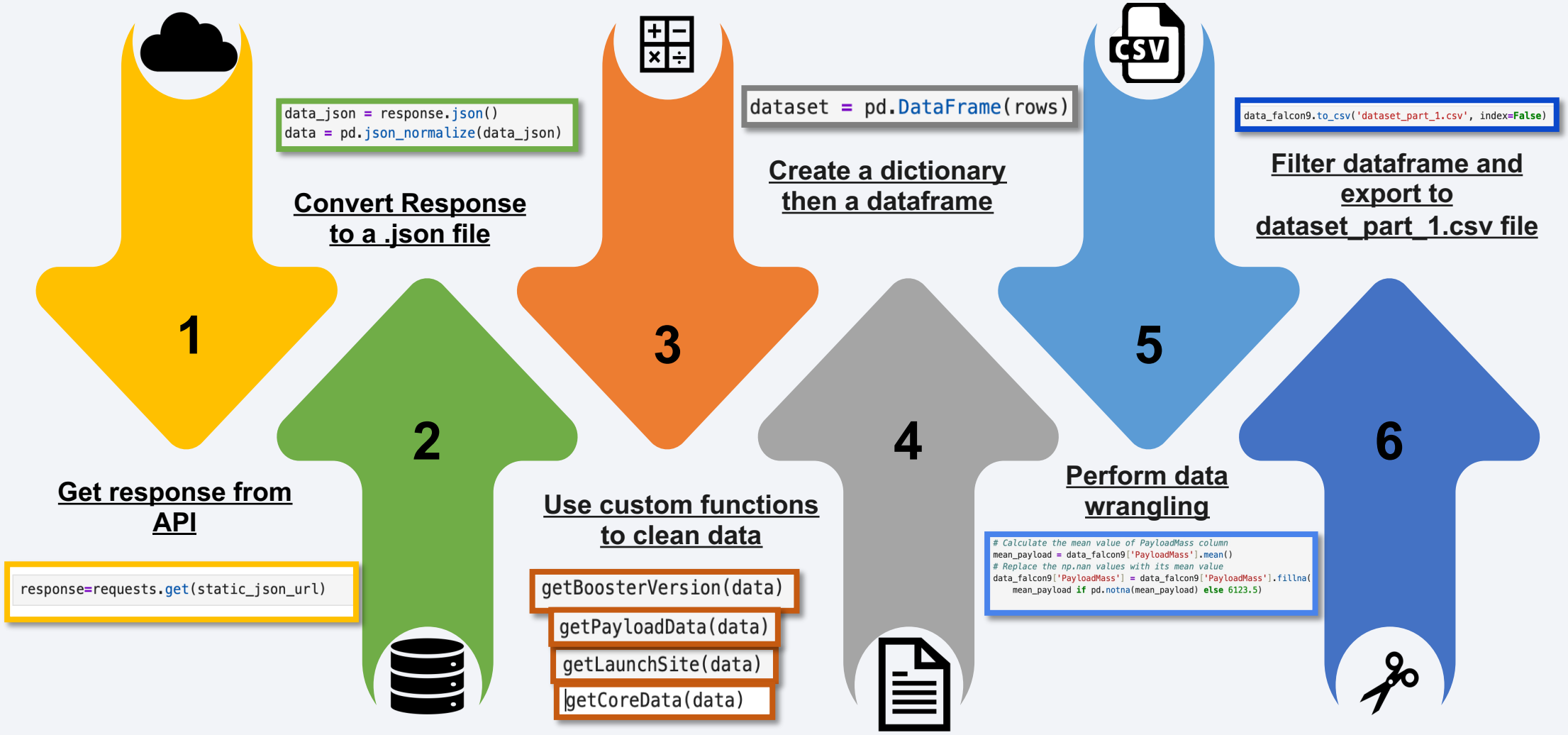
# Data Collection

Initially, data was collected from a Space X Application Programming Interface (API) using as JSON function to turn it into a data frame. Afterwards, the data was filtered to keep only Falcon 9 launches. Then, the data wrangling was performed to clean the information. Finally, the information was converted into a CSV file for future analysis.

The process is shown in the following diagram:



# Data Collection – SpaceX API




GitHub URL

# Data Collection - Scraping




Get Response from HTML

```
response = requests.get(static_url, headers=headers
```




Create BeautifulSoup Object

```
soup = BeautifulSoup(response.text, "html.parser")
```




Find web tables

```
html_tables= soup.find_all('table')
```




Get column names

```
for th in first_launch_table.find_all("th"):
    name = extract_column_from_header(th)
    if name is not None and len(name) > 0:
        column_names.append(name)
```




Create a dictionary and append data

```
df= pd.DataFrame({ key:pd.Series(value) for key, value in launch_dict.items() })
```




Convert dictionary to dataframe

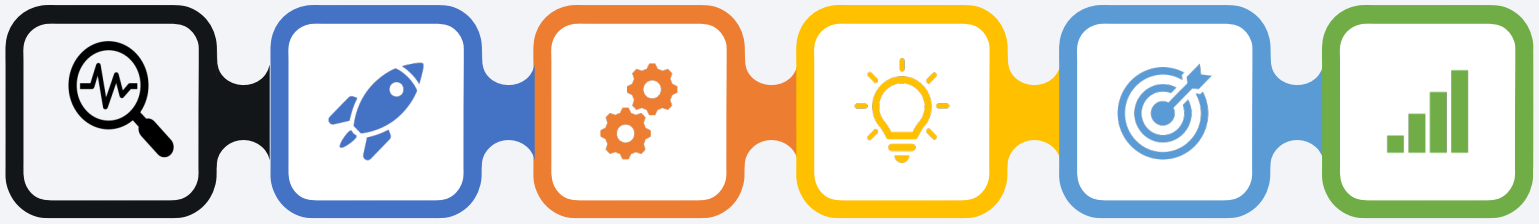
```
df.to_csv('spacex_web_scraped.csv', index=False)
```

GitHub URL

# Data Wrangling

Data wrangling was performed to find some patterns in the data, determine the success rate, and to get what would be the label for training supervised models. Moreover, those outcomes were converted into training labels with 1 means the booster successfully landed 0 means it was unsuccessful.

The process is depicted in the following chart:



**Step 1**

Exploratory Data Analysis EDA on dataset

**Step 2**

Calculate the number of launches per site

**Step 3**

Calculate the number and occurrence of each orbit

**Step 4**

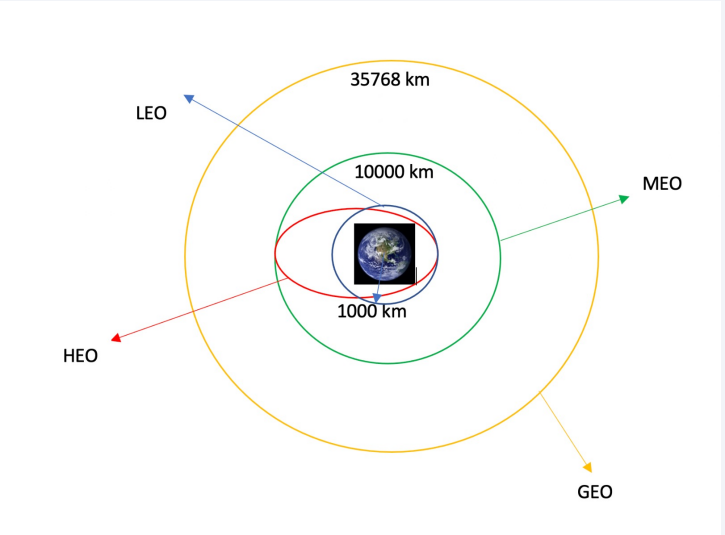
Calculate number and occurrence of mission outcome per orbit

**Step 5**

Determine the success rate

**Step 6**

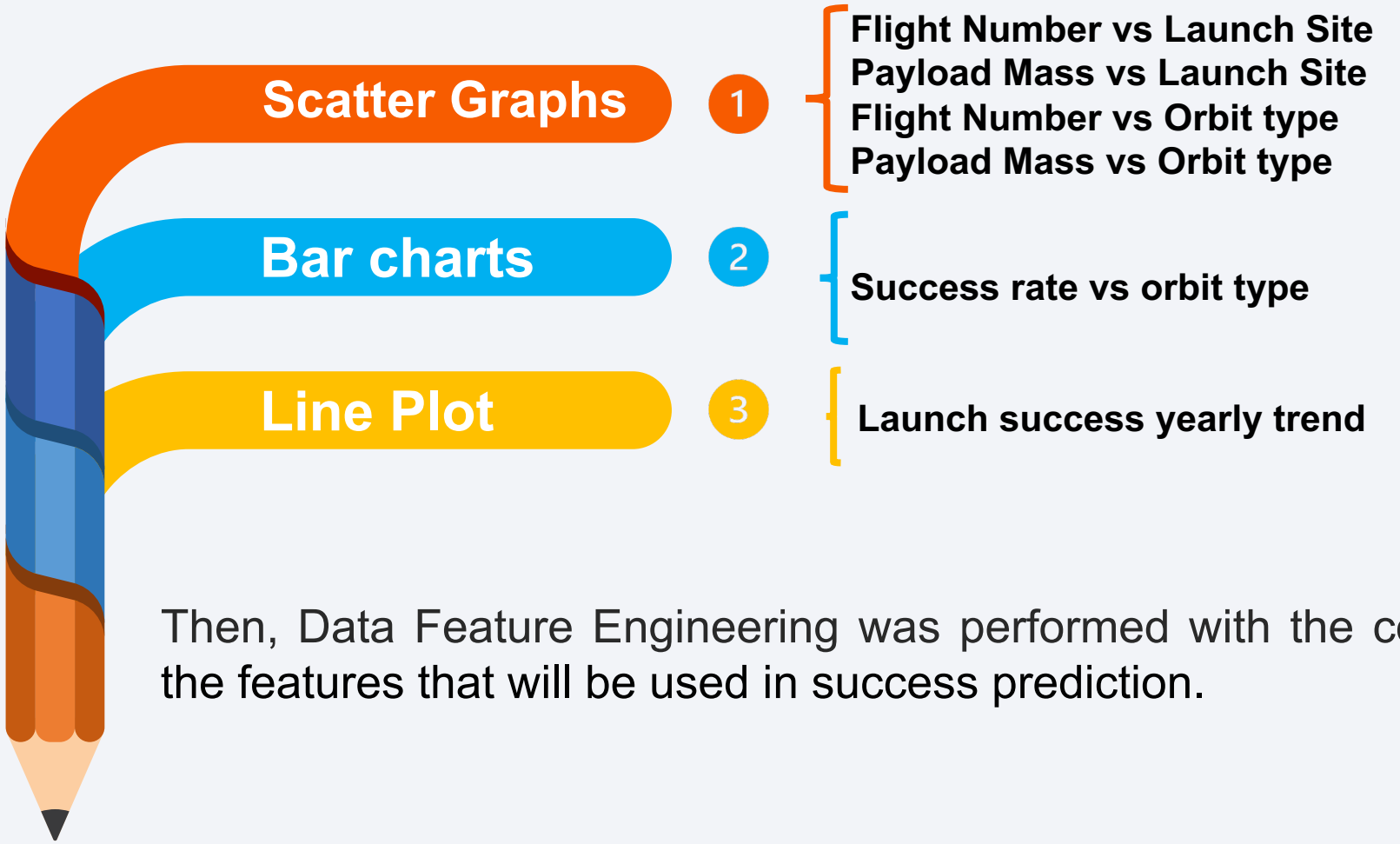
Export dataset as .CSV




GitHub URL

# EDA with Data Visualization

The exploratory data analysis was performed using Pandas and Matplotlib, using the following type of charts, which were helpful to observe graphically the data:




Then, Data Feature Engineering was performed with the collected data to select the features that will be used in success prediction.

GitHub URL

# EDA with SQL

In order to understand the Spacex DataSet, the dataset was loaded in a Db2 database, and the following SQL queries were executed:

01 **Display the names of the unique launch sites in the space mission.**

02 **Display 5 records where launch sites begin with the string 'CCA'.**

03 **Display the total payload mass carried by boosters launched by NASA (CRS).**

04 **Display average payload mass carried by booster version F9 v1.1.**

05 **List the date when the first successful landing outcome in ground pad was achieved.**

06 **List the names of the boosters which have success in drone ship and have payload mass between 4000 and 6000.**

07 **List the total number of successful and failure mission outcomes.**

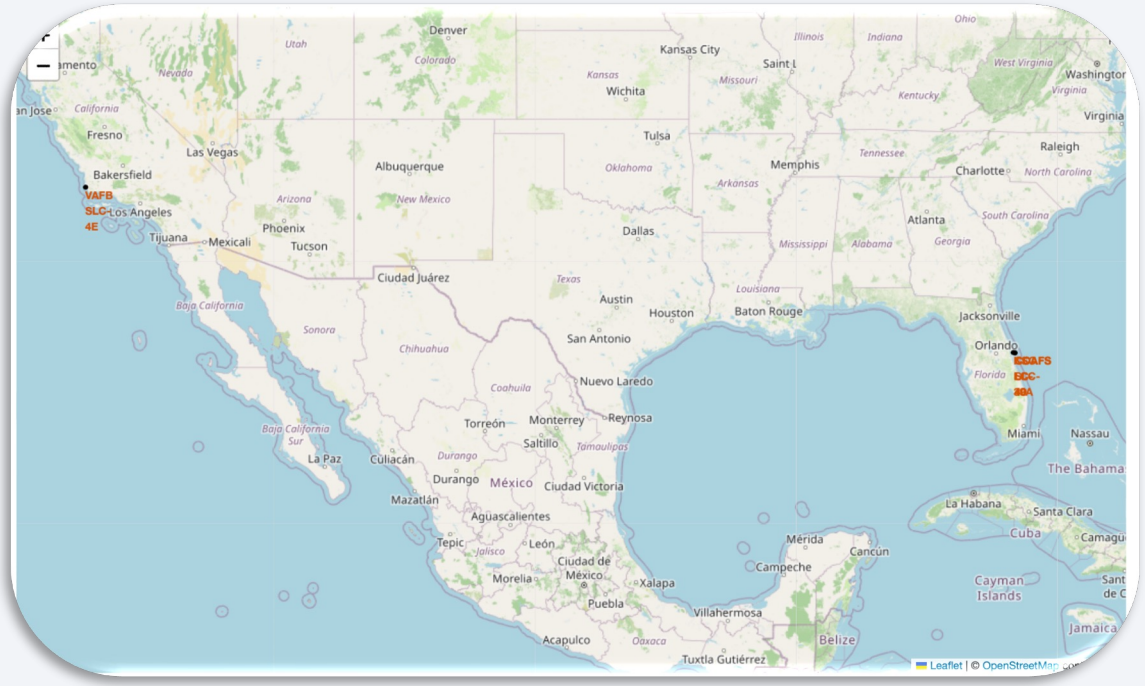
08 **List all the booster versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.**

09 **List the records which will display the month, failure landing outcomes in drone ship, booster versions, launch site for the months in 2015.**

10 **Rank the count of landing outcomes between the date 2010-06-04 and 2017-03-20, in descending order.¶**

GitHub URL

# Build an Interactive Map with Folium




- To visualize the Launch Data into an interactive map, the Latitude and Longitude Coordinates at each launch site were taken and added a Circle Marker around each launch site with a label of the name of the launch site

- These color-labeled markers, in marker clusters, allowed to easily identify which launch sites have relatively high success rates. A green marker for a positive launch and ared marker  failed launch were used.

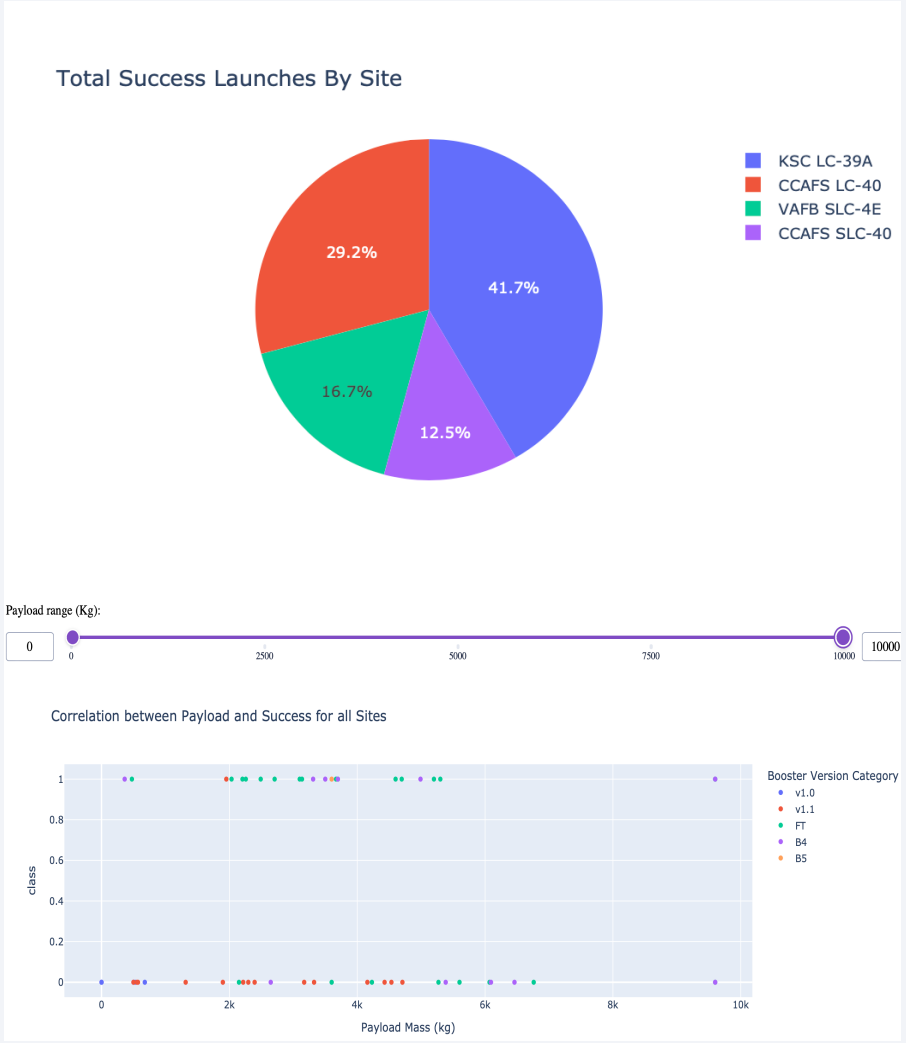
- Afterwards, using a defined function, called CALCULATE_DISTANCE,  the distance between the launch site and other features like railway, highway, coastline, and a town, were measured.

- Finally, a line with a label including the distances were added.

GitHub URL

# Build a Dashboard with Plotly Dash

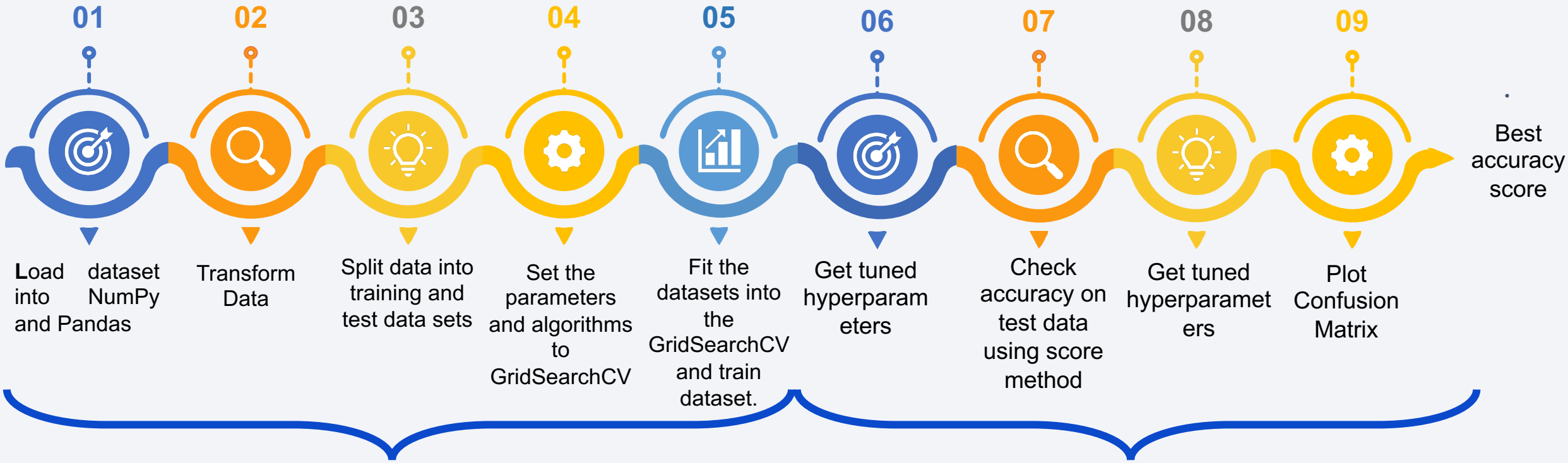
- The designed dashboard application contains input components such as a dropdown list and a range slider to interact with a pie chart and a scatter point chart.
- On the dropdown list there are four different launch sites, once selected, it is possible to choose one specific site and check a pie chart visualizing launch success counts.
- Additionally, a scatter plot, with the x axis as the payload and the y axis as the launch outcome, is set to visualize how payload may be correlated with mission outcomes for selected site. Moreover, the Booster version is color-label on each scatter point so that it is easily to observe mission outcomes.
- Finally, the range slider helps to select a payload to find if this variable is correlated to mission outcome and identify some visual patterns.




GitHub URL

# Predictive Analysis (Classification)

To built, evaluate, improve, and found the best performing classification model (logistic regression, support vector machine, decision tree classifier, k-nearest neighbors), the following process was followed:




GitHub URL

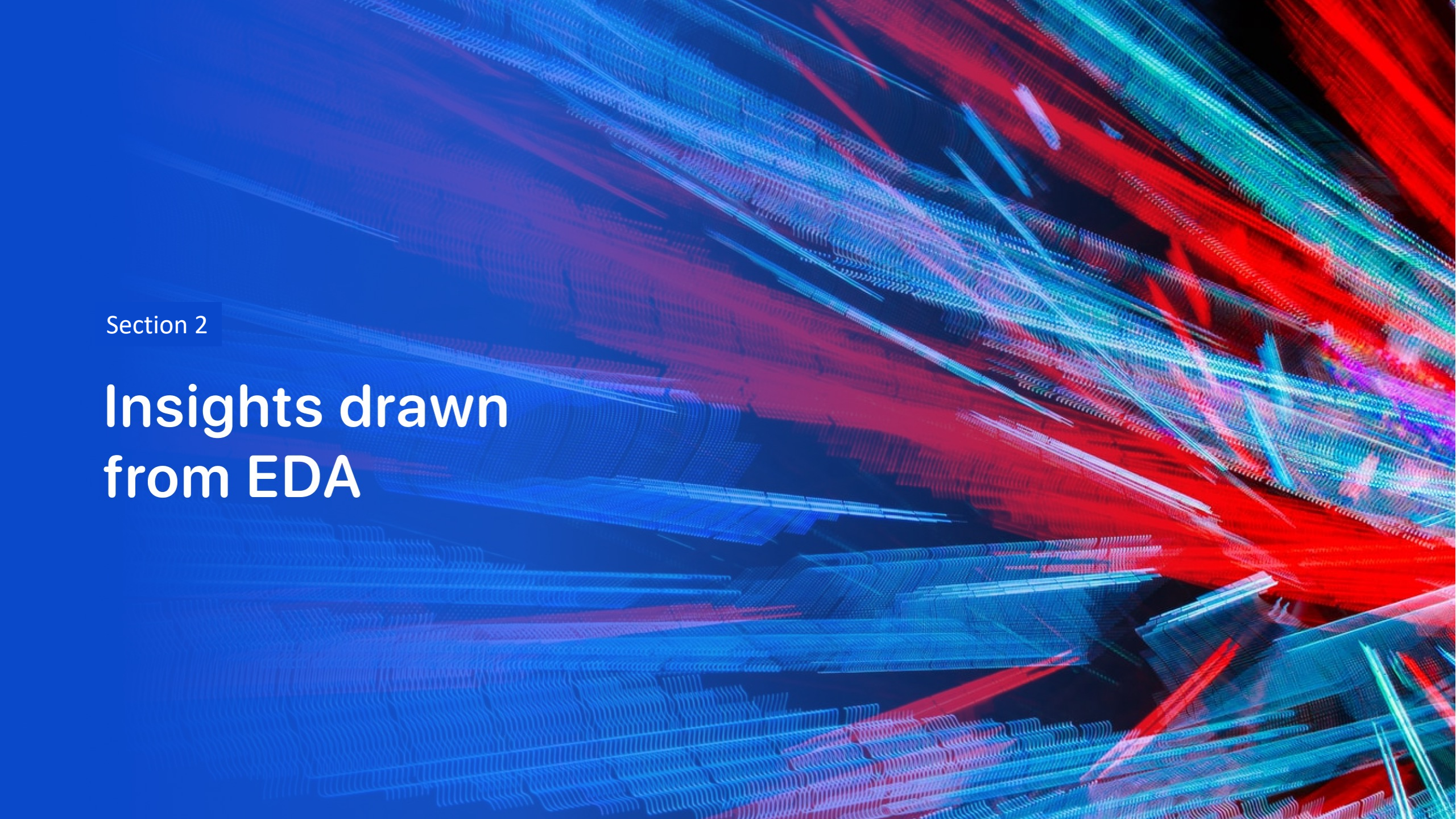
# Results

Results are split  into the following  sections:

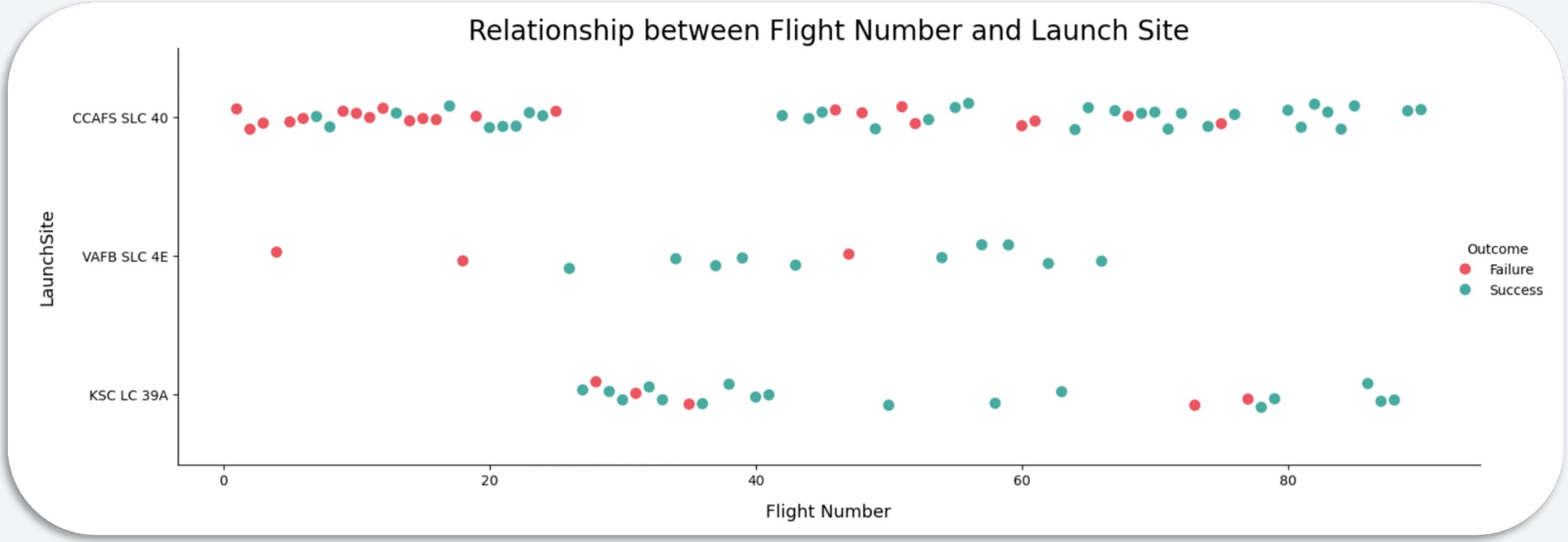
- EDA with SQL
- Matplotlib and Seaborn EDA with Visualization (Matplotlib and Seaborn)
- Maps features visualization with Folium
- Dash board visual analysis
- Predictive Analysis

Section 2

# Insights drawn from EDA

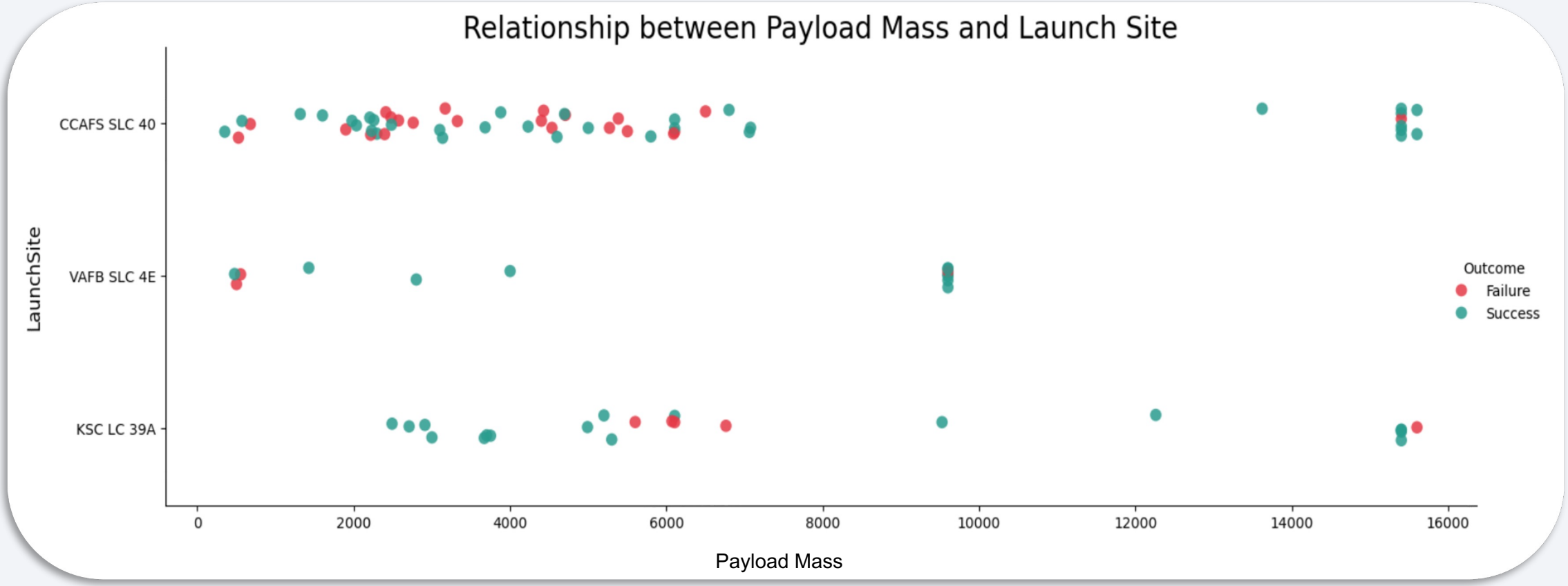


# Flight Number vs. Launch Site



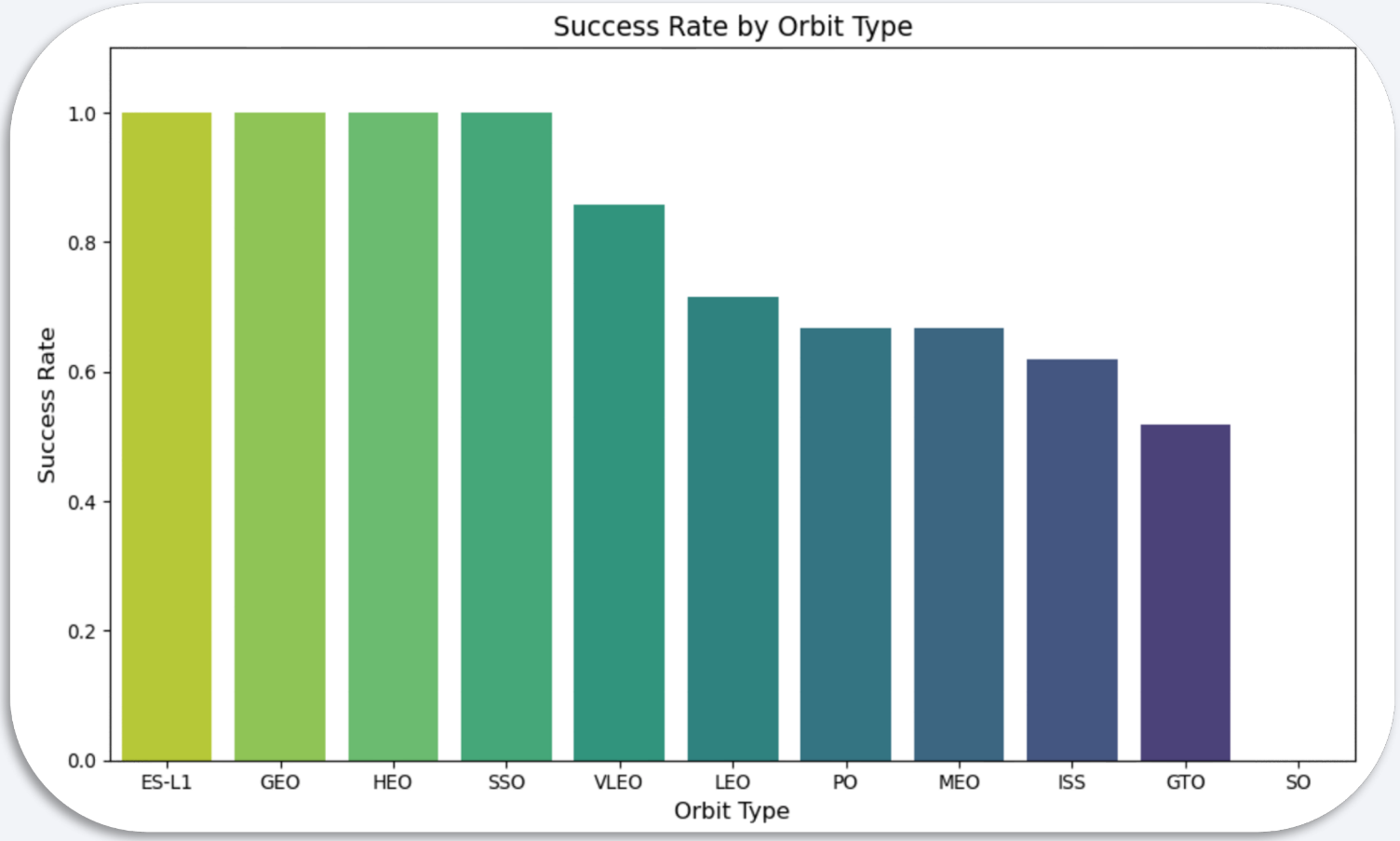

It is evident that as the flight number the number increases, the rate of successful mission increases proportionally.

# Payload vs. Launch Site



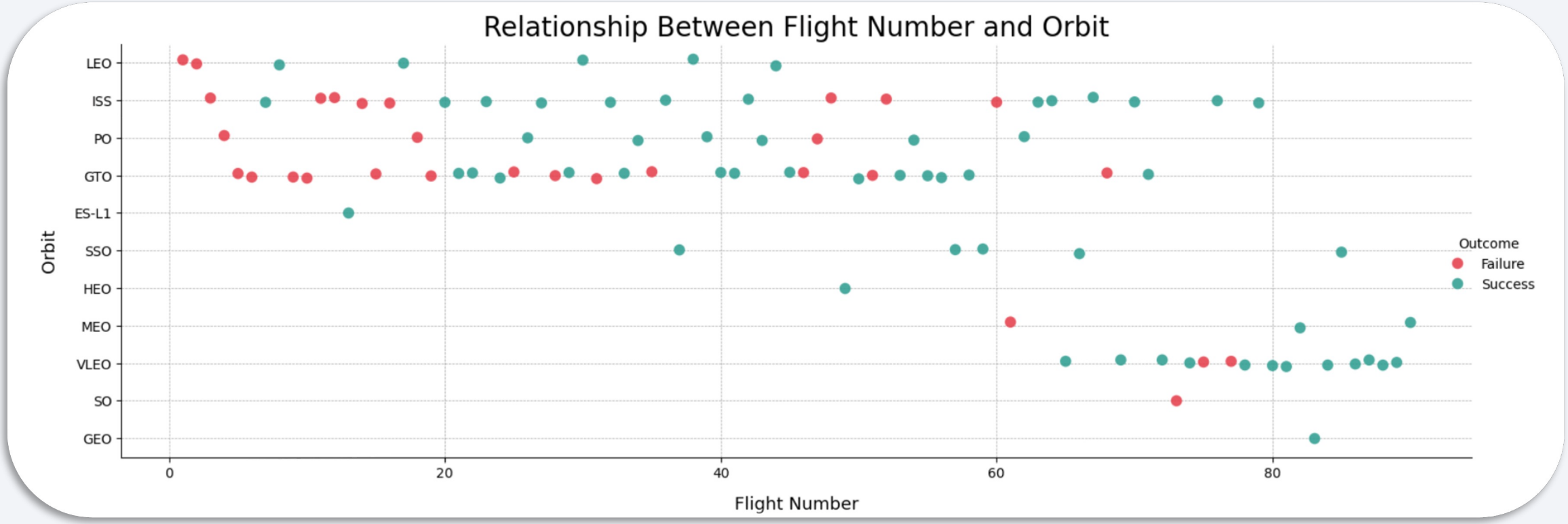

It is spotted that for the VAFB-SLC launch site there are no rockets launched for payload mass greater than 10000. Furthermore, with heavier payloads the rate of failure mission decreases.

# Success Rate vs. Orbit Type



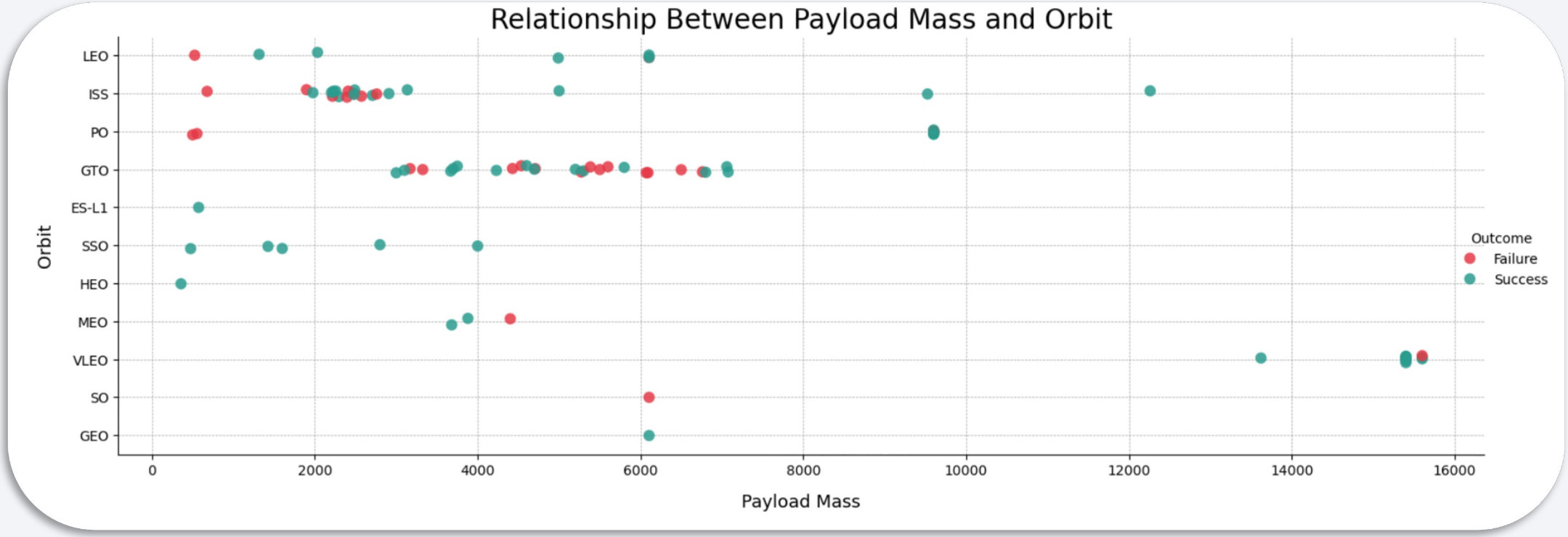

While rockets launched into orbits ES-L1, GEO, HEO, and SSO had 100% of success, the rockets put into the rest of orbits showed an average of 60 % rate of success

# Flight Number vs. Orbit Type



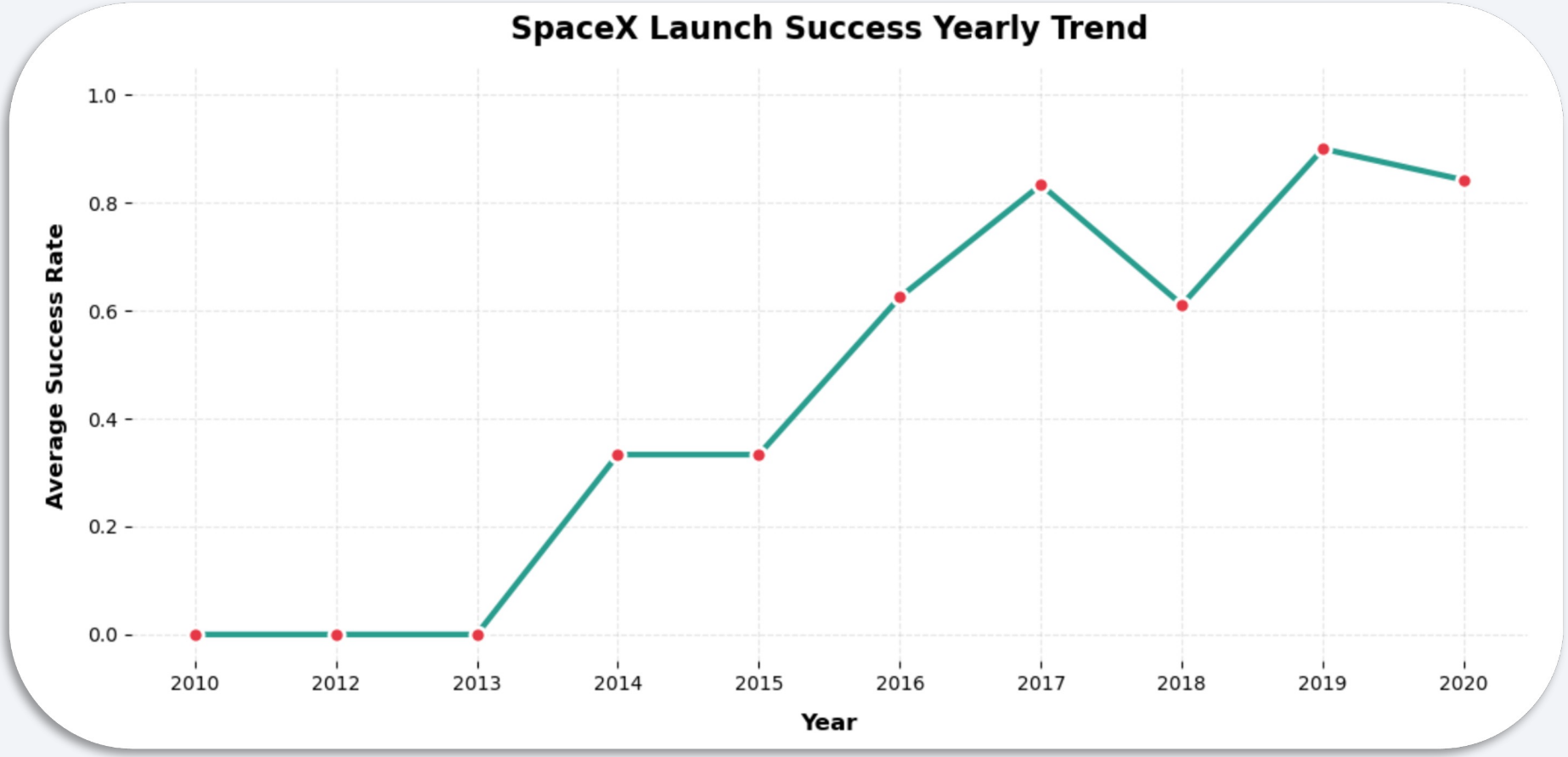

Overall, with the increase in the flight number the rate of success increases. Moreover, SSO, HEO, MEO, VLEO, SO, and GEO orbits were used mostly after flight number 40. Additionally, while in the LEO orbit success seems to be related to the number of flights, in the GTO orbit, there appears to be no relationship between flight number and success.

# Payload vs. Orbit Type




While launches into ISS, PO, and VLEO orbits are more successful at a higher payload mass, launches into SSO are entirely positive with a lower payload mass. However, for launches into GTO orbit, there is apparently no relation between successful and unsuccessful landings regarding the payload mass.

# Launch Success Yearly Trend




Initially, in the first three years the success rate is zero. However, after 2013, the positive outcome kept increasing till 2020, reaching its peak in 2019 with nearly 90 percent

# All Launch Site Names

**SQL QUERY**

```
SELECT DISTINCT LAUNCH_SITE FROM TBLSPACEX
```

**EXPLANATION**

The command ***DISTINCT*** was used in the query to show only Unique values in the ***Launch_Site*** column from ***tblSpace***

| Launch_Site |
| --- |
| CCAFS LC-40 |
| VAFB SLC-4E |
| KSC LC-39A |
| CCAFS SLC-40 |

# Launch Site Names Begin with 'CCA'

| Date | Time (UTC) | Booster_Version | Launch_Site | Payload | PAYLOAD_MASS__KG_ | Orbit | Customer | Mission_Outcome | Landing_Outcome |
|---|---|---|---|---|---|---|---|---|---|
| 2010-06-04 | 18:45:00 | F9 v1.0 B0003 | CCAFS LC-40 | Dragon Spacecraft Qualification Unit | 0 | LEO | SpaceX | Success | Failure (parachute) |
| 2010-12-08 | 15:43:00 | F9 v1.0 B0004 | CCAFS LC-40 | Dragon demo flight C1, two CubeSats, barrel of Brouere cheese | 0 | LEO (ISS) | NASA (COTS) NRO | Success | Failure (parachute) |
| 2012-05-22 | 7:44:00 | F9 v1.0 B0005 | CCAFS LC-40 | Dragon demo flight C2 | 525 | LEO (ISS) | NASA (COTS) | Success | No attempt |
| 2012-10-08 | 0:35:00 | F9 v1.0 B0006 | CCAFS LC-40 | SpaceX CRS-1 | 500 | LEO (ISS) | NASA (CRS) | Success | No attempt |
| 2013-03-01 | 15:10:00 | F9 v1.0 B0007 | CCAFS LC-40 | SpaceX CRS-2 | 677 | LEO (ISS) | NASA (CRS) | Success | No attempt |

**SQL QUERY**

```
SELECT * FROM SPACEXTBL WHERE LAUNCH_SITE LIKE 'CCA%' LIMIT 5;
```

**EXPLANATION**

The command ***LIKE*** was used in the query to show values in the column launch_site which start with the letters CCA.

# Total Payload Mass

| TOTAL_PAYLOAD_MASS |
| --- |
| 45596 |

**SQL QUERY**

```
SELECT SUM(PAYLOAD_MASS__KG_) AS TOTAL_PAYLOAD_MASS FROM SPACEXTBL
WHERE CUSTOMER = 'NASA (CRS)'
```

**EXPLANATION**

The commands ***SUM and WHERE*** were used in the query to sum values of the column launch_site equal to 'NASA (CRS)'.

# Average Payload Mass by F9 v1.1

| AVG_MASS_F9_v11 |
| --- |
| 2534.6666666666665 |

**SQL QUERY**

```
SELECT AVG(PAYLOAD_MASS__KG_) AS AVG_MASS_F9_V11 FROM SPACEXTBL WHERE BOOSTER_VERSION LIKE 'F9 V1.1%';
```

**EXPLANATION**

The command ***AVG*** calculates the average in the column ***PAYLOAD_MASS_KG_,*** and the command ***WHERE*** filters the dataset to only perform calculations on ***Booster_version F9 v1.1***

# First Successful Ground Landing Date

| MIN(date) |
| --- |
| 2015-12-22 |

**SQL QUERY**

```
SELECT MIN(DATE) FROM SPACEXTBL WHERE LANDING_OUTCOME = 'SUCCESS (GROUND PAD)';
```

**EXPLANATION**

The command ***MIN(date)*** calculates the first successful landing outcome on ground pad.

# Successful Drone Ship Landing with Payload between 4000 and 6000

## SQL QUERY

```
SELECT BOOSTER_VERSION FROM SPACEXTBL
WHERE LANDING_OUTCOME LIKE '%DRONE SHIP%'
AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000;
```

## EXPLANATION

The commands ***LIKE, AND, BETWEEN*** get the list boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000.

| Booster_Version |
|---|
| F9 FT B1020 |
| F9 FT B1022 |
| F9 FT B1026 |
| F9 FT B1021.2 |
| F9 FT B1031.2 |

# Total Number of Successful and Failure Mission Outcomes

## SQL QUERY

```
SELECT CASE
      WHEN MISSION_OUTCOME LIKE '%UCCESS%'
THEN 'SUCCESS'
      WHEN MISSION_OUTCOME LIKE '%AILURE%'
THEN 'FAILURE'
   END AS OUTCOME_TYPE,
   COUNT(*) AS TOTAL
FROM SPACEXTBL
WHERE MISSION_OUTCOME LIKE '%UCCESS%' OR
MISSION_OUTCOME LIKE '%AILURE%'
GROUP BY OUTCOME_TYPE;
```

| Outcome_Type | Total |
|---|---|
| Failure | 1 |
| Success | 100 |

## EXPLANATION

The command ***CASE*** and a subquery were used to calculate the total number of successful and failure mission outcomes, grouped in two rows.

# Boosters Carried Maximum Payload

**SQL QUERY**

```
SELECT DISTINCT BOOSTER_VERSION FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL);
```

**EXPLANATION**

The command ***DISTINCT*** and a subquery were used to obtain a List with the names of the booster which have carried the maximum payload mass.

| Booster_Version | PAYLOAD_MASS__KG_ |
|---|---|
| F9 B5 B1048.4 | 15600 |
| F9 B5 B1049.4 | 15600 |
| F9 B5 B1051.3 | 15600 |
| F9 B5 B1056.4 | 15600 |
| F9 B5 B1048.5 | 15600 |
| F9 B5 B1051.4 | 15600 |
| F9 B5 B1049.5 | 15600 |
| F9 B5 B1060.2 | 15600 |
| F9 B5 B1058.3 | 15600 |
| F9 B5 B1051.6 | 15600 |
| F9 B5 B1060.3 | 15600 |
| F9 B5 B1049.7 | 15600 |

# 2015 Launch Records

| Month | Landing_Outcome | Booster_Version | Launch_Site |
|---|---|---|---|
| 01 | Failure (drone ship) | F9 v1.1 B1012 | CCAFS LC-40 |
| 04 | Failure (drone ship) | F9 v1.1 B1015 | CCAFS LC-40 |

**SQL QUERY**

```
SELECT SUBSTR(DATE, 6, 2) AS MONTH, LANDING_OUTCOME, BOOSTER_VERSION,
LAUNCH_SITE FROM SPACEXTBL WHERE SUBSTR(DATE, 0, 5) = '2015' AND
LANDING_OUTCOME = 'FAILURE (DRONE SHIP)';
```

**EXPLANATION**

The command ***SUBSTR(),*** was used to get a list of the failed landing outcomes in drone ship, their booster versions, and launch site names for in year 2015 in one table.

# Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

**SQL QUERY**

```
SELECT LANDING_OUTCOME, COUNT(*) AS OUTCOME_COUNT FROM SPACEXTBL WHERE DATE BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY LANDING_OUTCOME ORDER BY OUTCOME_COUNT DESC;
```

**EXPLANATION**

Commands ***COUNT, WHERE*** and ***ORDER BY,*** were use to Rank the count of landing outcomes, such as Failure (drone ship) or Success (ground pad), between the date 2010-06-04 and 2017-03-20, in descending order.

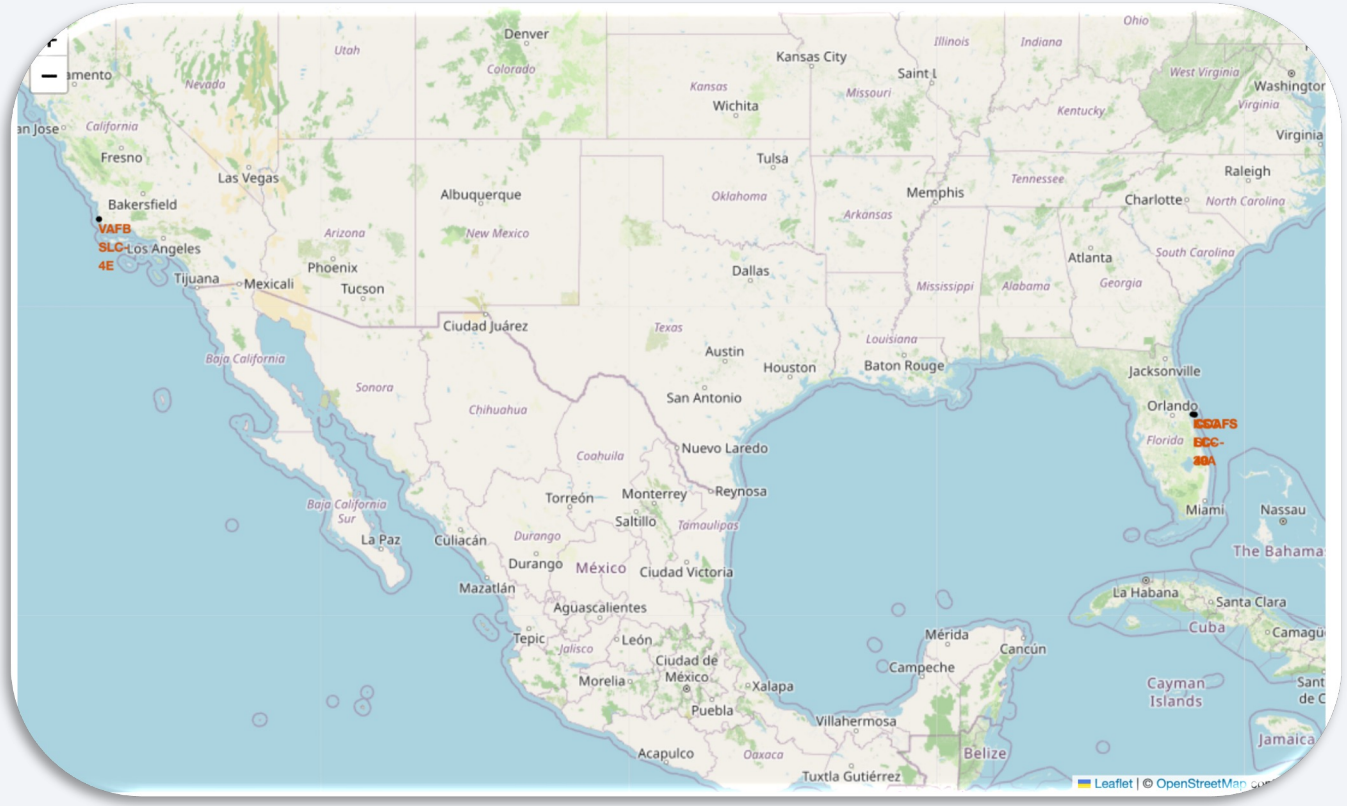
| Landing_Outcome | Outcome_Count |
|---|---|
| No attempt | 10 |
| Success (drone ship) | 5 |
| Failure (drone ship) | 5 |
| Success (ground pad) | 3 |
| Controlled (ocean) | 3 |
| Uncontrolled (ocean) | 2 |
| Failure (parachute) | 2 |
| Precluded (drone ship) | 1 |



Section 3

# Launch Sites Proximities Analysis

# Launch Sites Map

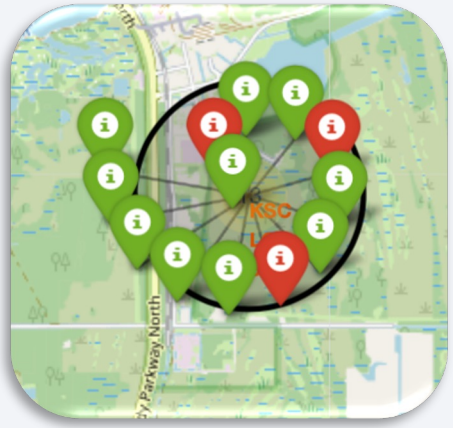



In this global map the four launch sites are plotted. While the VAFB SLC-4E site is located on the west coast of the EE.UU., the three others : KSC LC-39A, CCAF SLC-40, and CCAFS LC-40, are located in the east coast of the same country.

# Launch Outcomes Grouped by Site




VAFB SLC-4E




KSC LC-39A



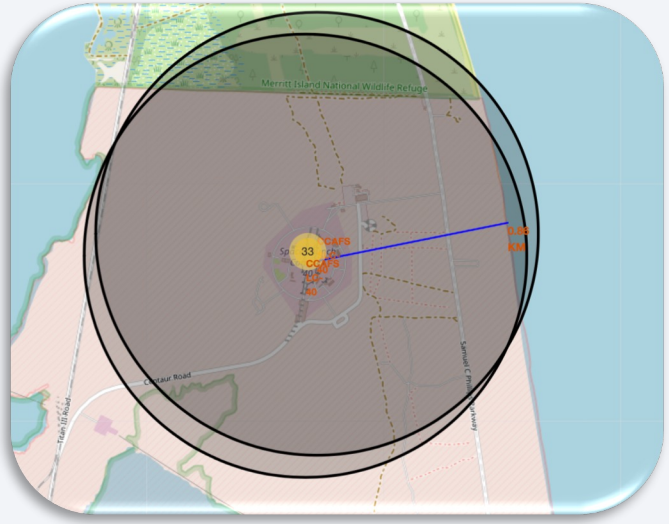

CCAFS LC-40



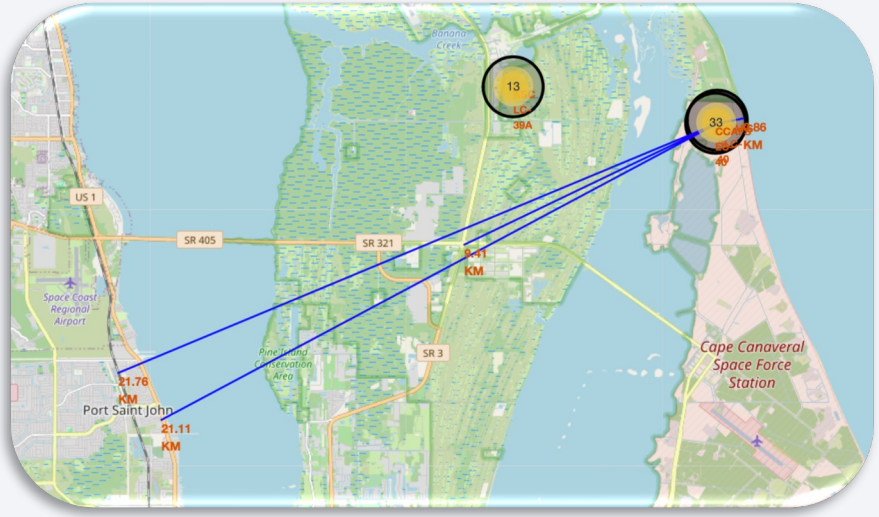

CCAF SLC-40

With the ***MASTER_CLUSTER()*** tool, the launch outcomes were grouped and plotted in the four sites. Additionally, markers were used to highlight the positive (green) and negative (red) results.

# Proximity Analysis



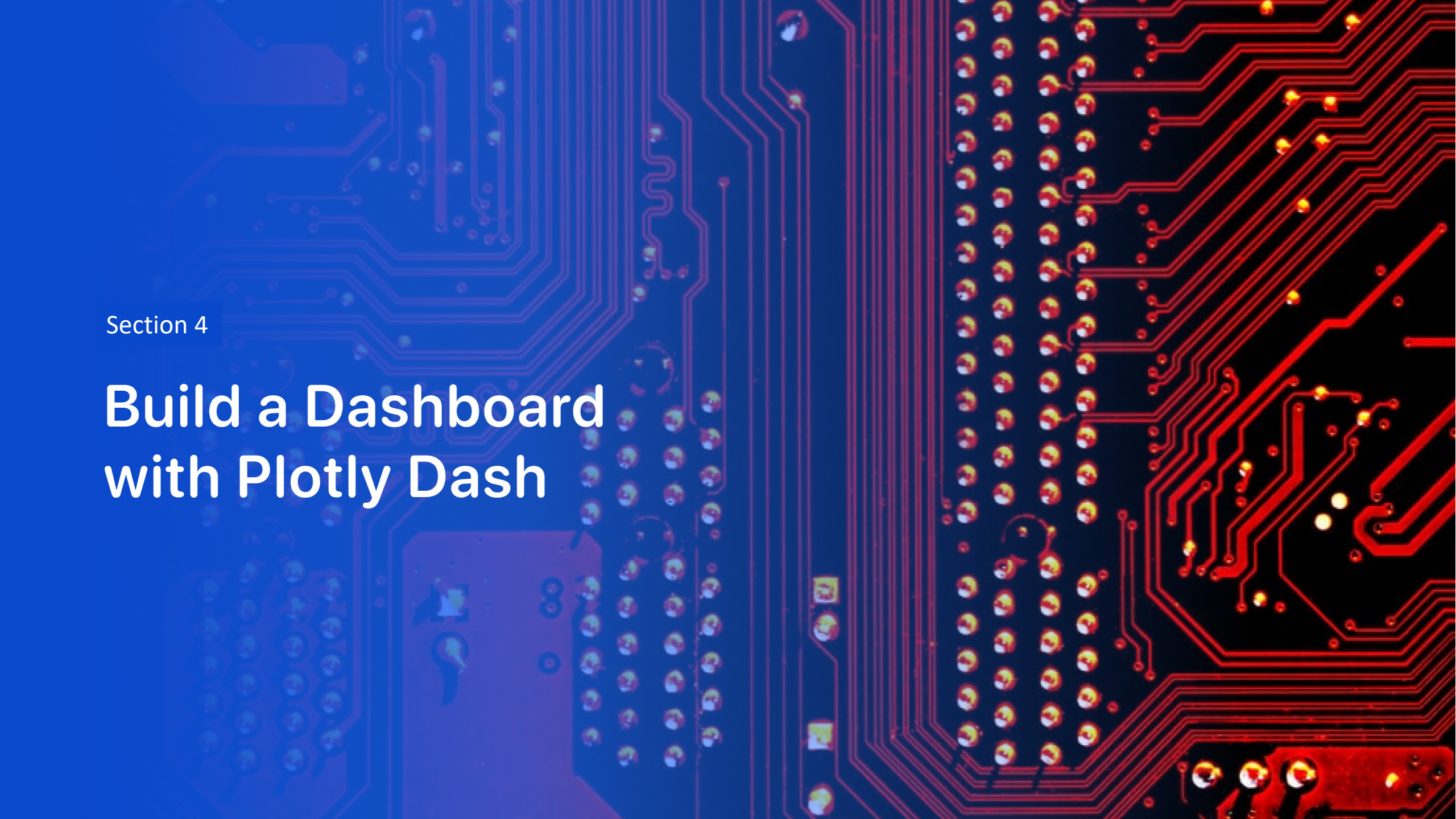

Distance to the coast.




Distance to selected features.

The CCAFS LC-40 site was choosen to perform an analysis of its proximities such as railway, highway, coastline, and a town, with distance calculated and displayed.

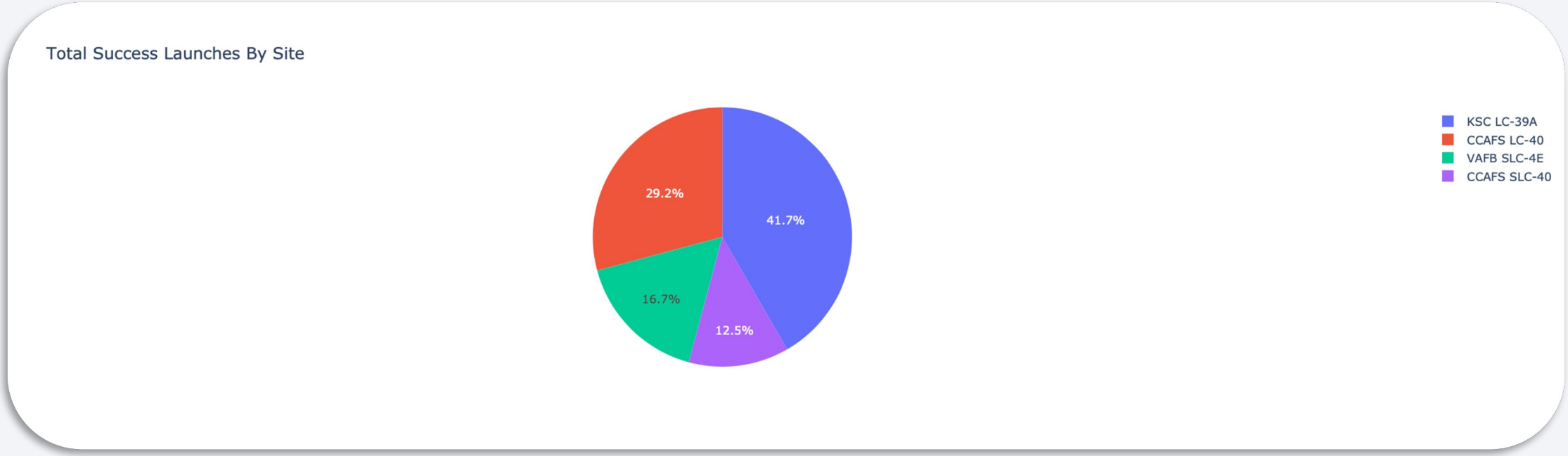
Distance to the coast: 0.85 km
Distance to the closest highway: 9.41 km
Distance to the closest railway: 21.76 km
Distance to the closest town: 21.11 km



Section 4

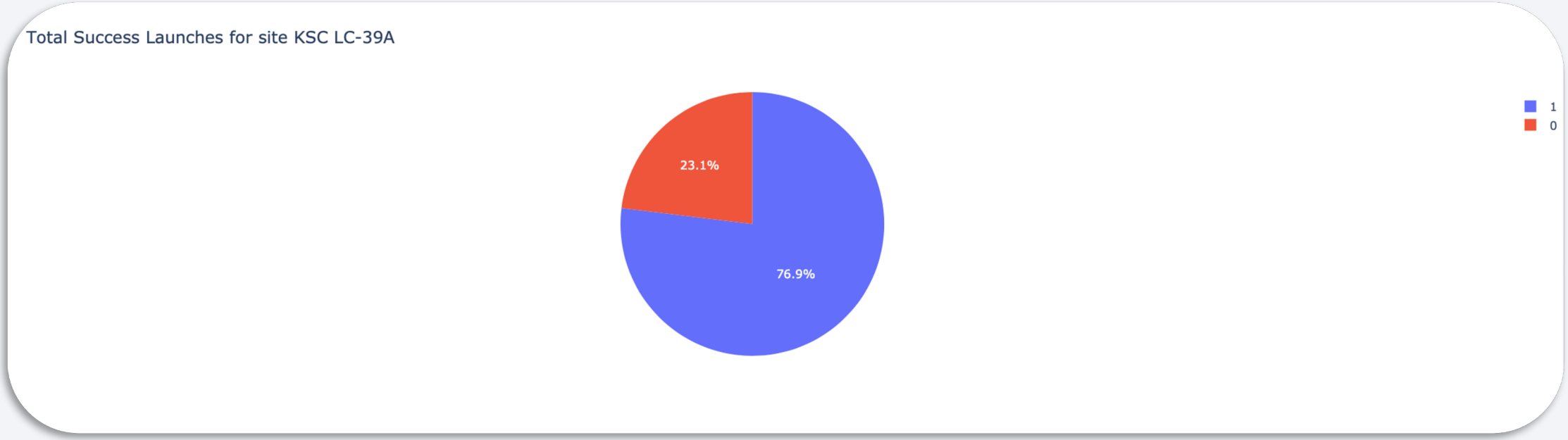
# Build a Dashboard with Plotly Dash

# Total Success Launches by Site




This portion of the dashboard depicts a pie chart which is being used to show the rate of success launches per site. Additionally, by using the dropdown list it is possible to select each site in order to check the success rate per each one.

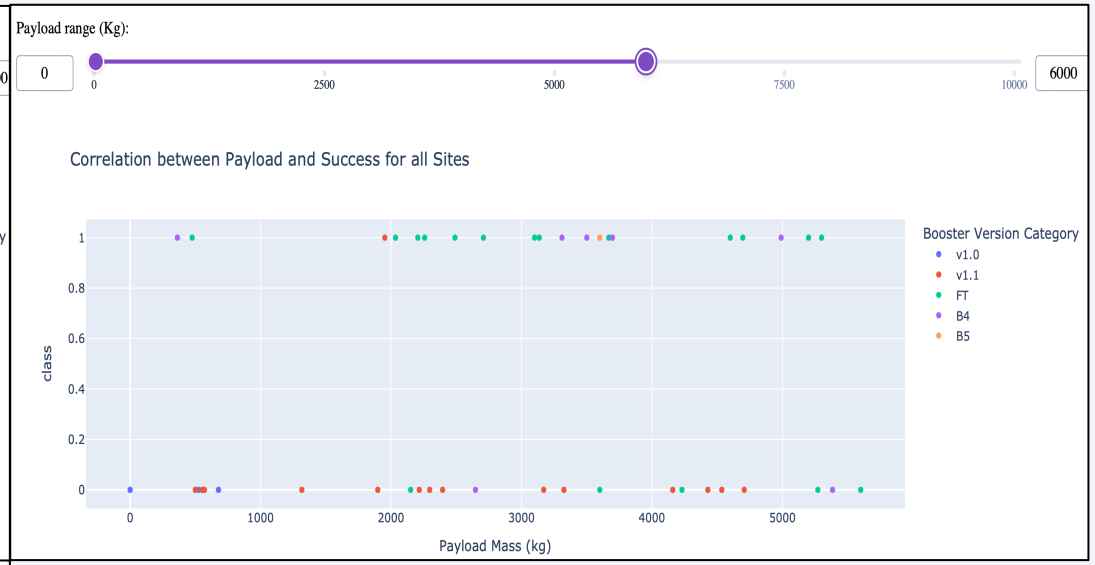
# Launch Site with Highest Launch Success Ratio

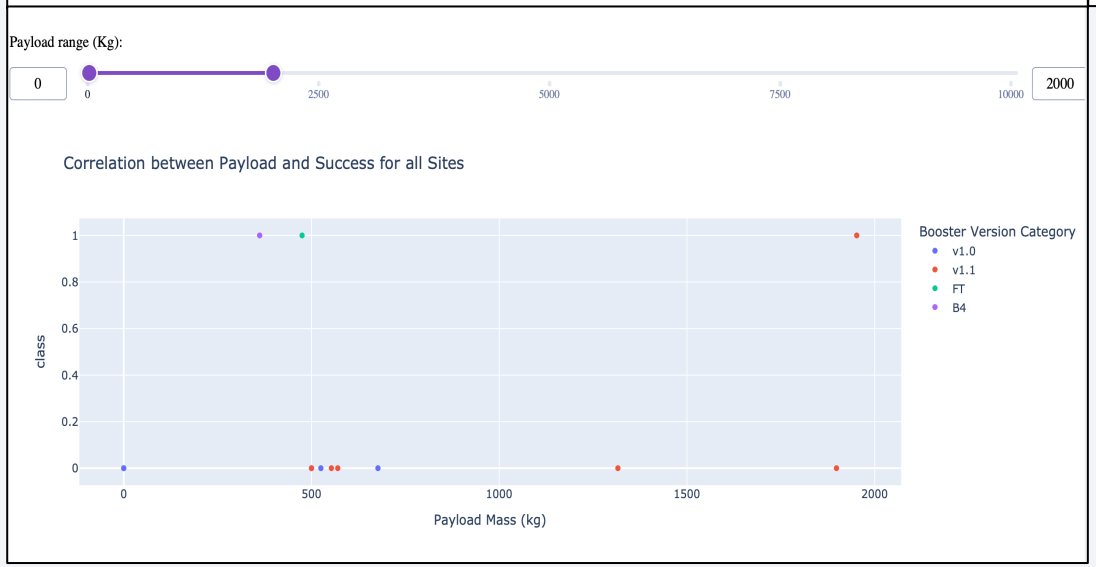



The pie chart on the dashboard depicts the success rate of the KSC LC-39A site. On one hand, launches made from this site achieved a 76.9% success rate (blue). On the other hand, they had a 23.1% failure rate (red)

# Correlation Between Payload and Launch Outcome



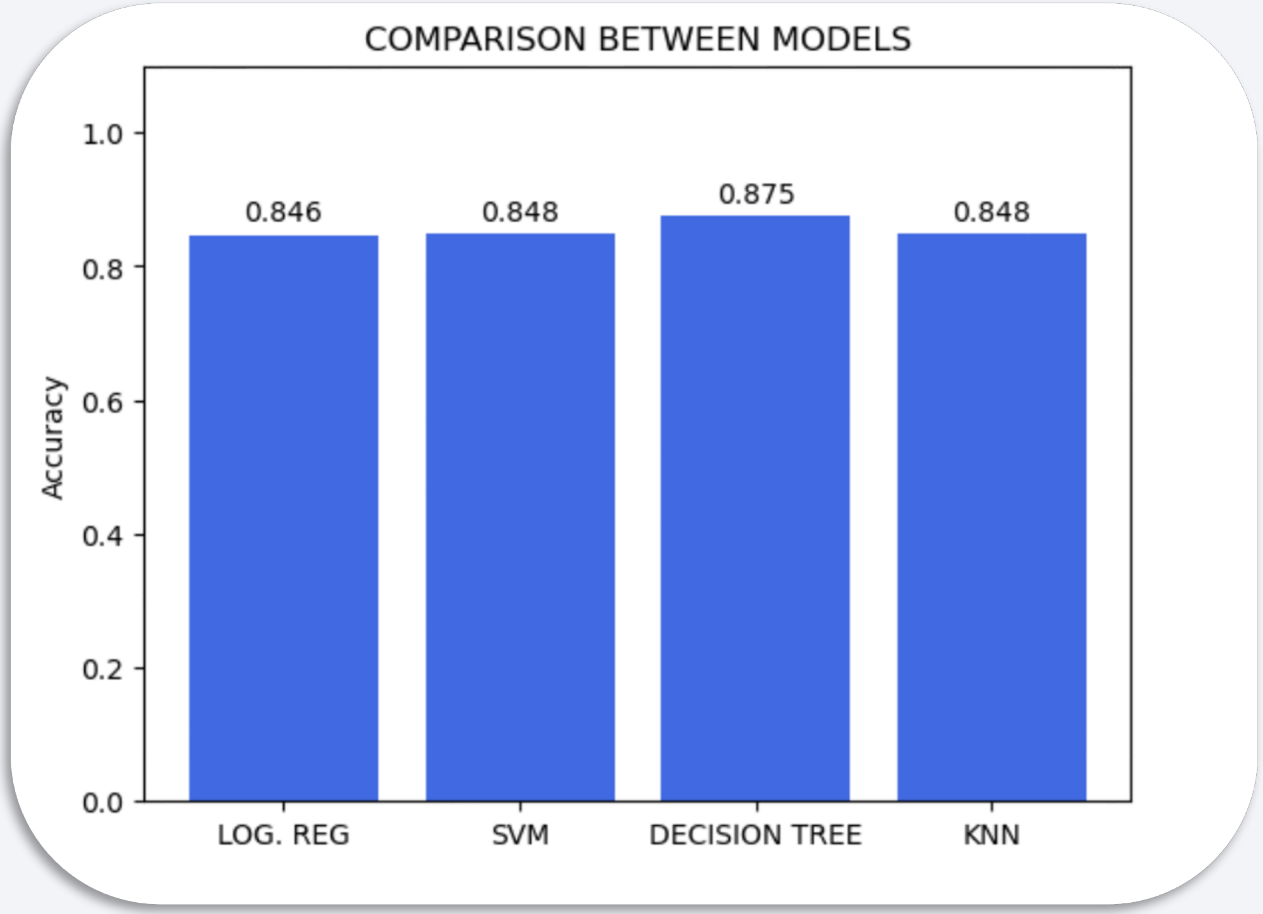

In these pictures, it is evident how the success rate decreases proportionally with the reduction of the payload. Additionally, the FT Booster shows lower failure rate in all payload ranges among the other boosters.

Section 5

# Predictive Analysis (Classification)

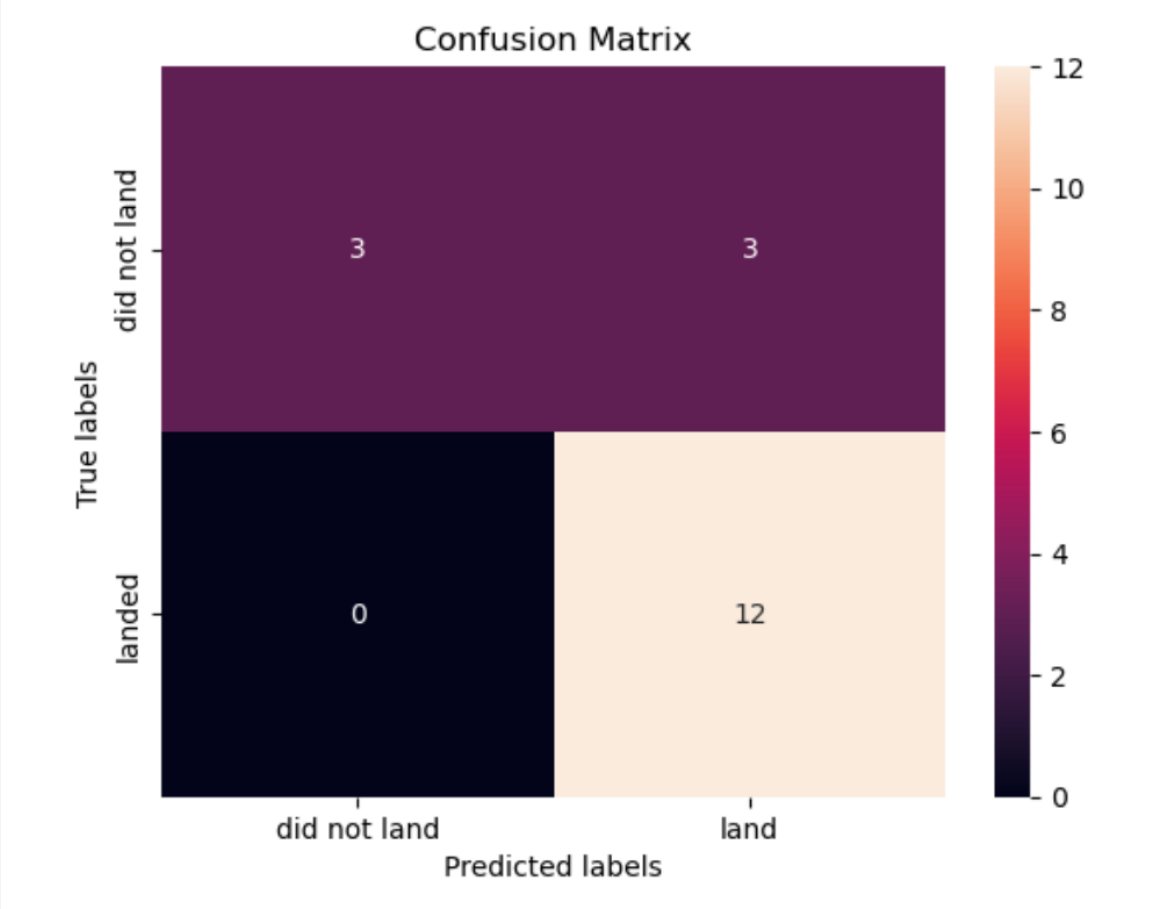


# Classification Accuracy




Instead of using the score parameter, which is fairly similar across all four models (about 0.833), the best_score_attribute was chosen to select the decision tree, which yielded the highest value: 0.875.

# Decision Tree Confusion Matrix




- **Perfect Landing Recall:** 100 % accuracy in detecting successful landings (0 false negatives).

- **High Misclassification on Failures:** 50 % of "did not land" cases (3 of 6) were incorrectly predicted as successful landings.

- **83.3 % Overall Accuracy:** Correctly predicted 15 out of 18 total samples, with a class imbalance toward successful landings (12 vs. 6).

# Conclusions

- Rockets launched into ES-L1, GEO, HEO, and SSO orbits had a 100% success rate.
- With a 76.9% success rate, the KSC LC-39A site had the highest launch success rate among all sites.
- Overall, the decision tree model achieved the best performance in predicting successful Falcon 9 landings.
- With heavier payloads, the mission failure rate decreases.
- The launch success rate increased from 2013 to 2020.

# Thank you!

